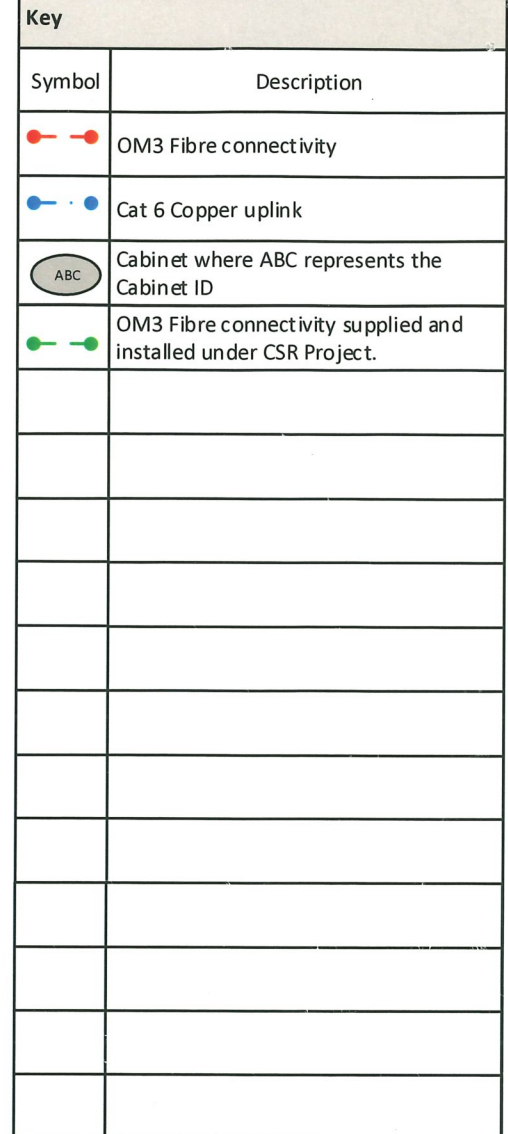


General Notes

- ## Installation Notes



Contact: Carla Smith

Designer: MaryAnne Malaluan

[illegible]

File Name: [School Initials-Ministry of Education #-Job #, Rev]

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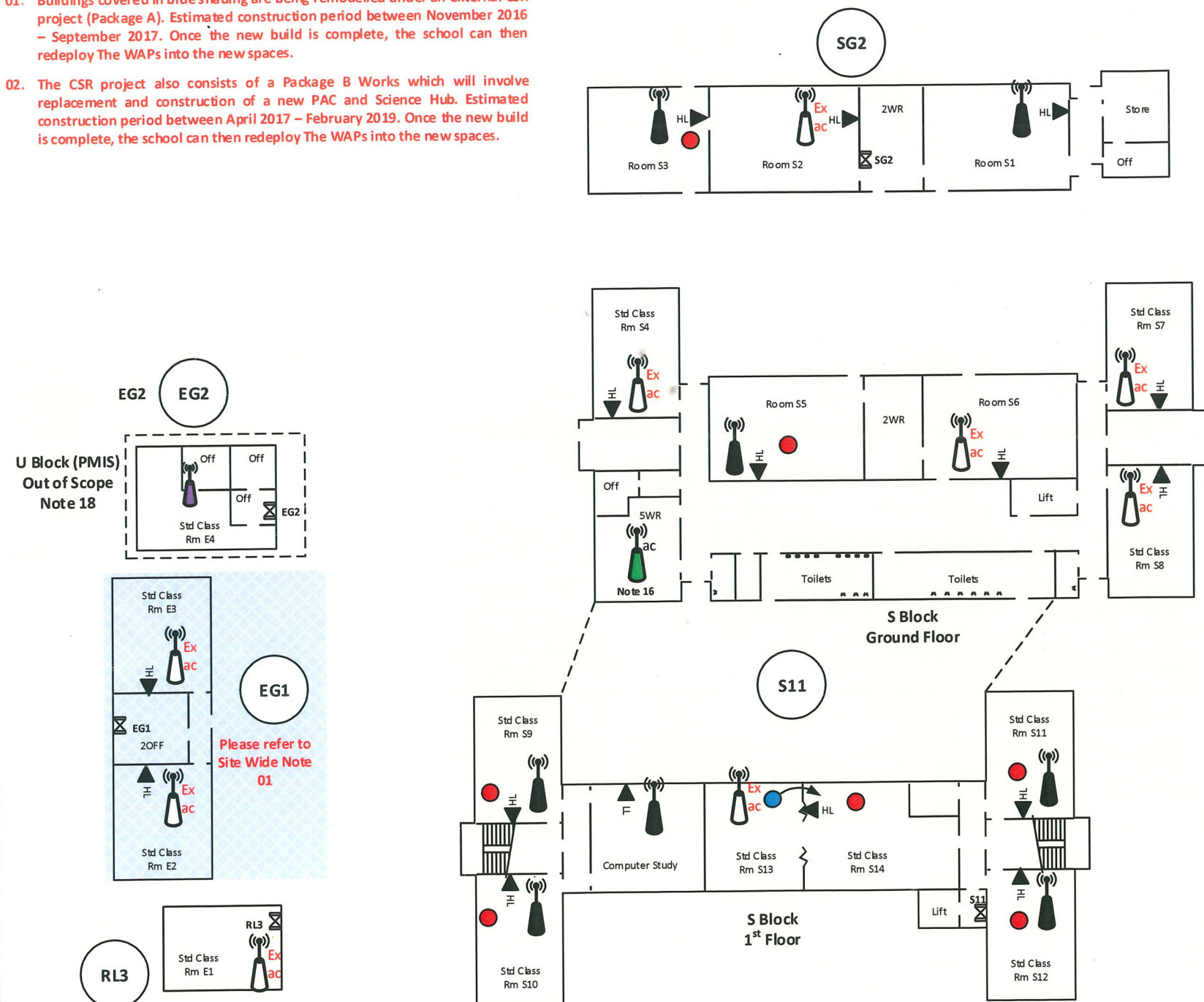
WIRELESS ACCESS POINT (WAP) REQUIREMENTS

General Notes (All Notes Apply)

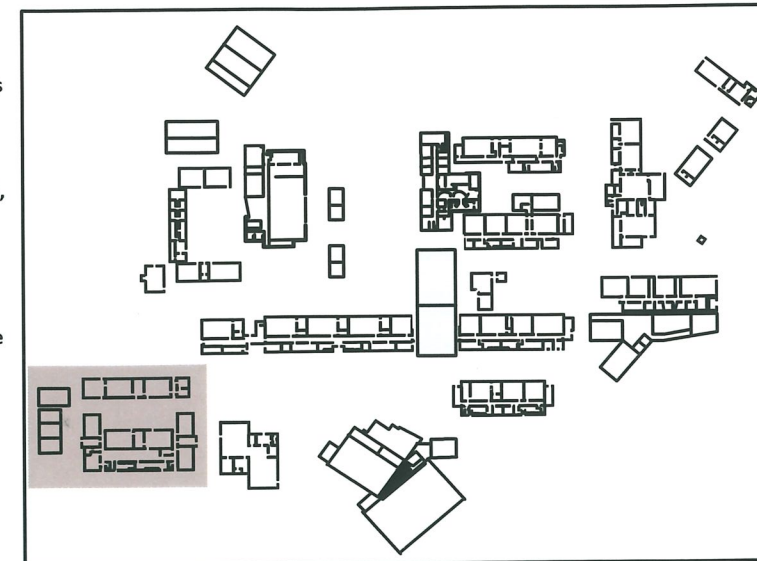
01. The plans below show the proposed location of WAPs. Do not mount WAPs back to back in adjacent classrooms, within 300mm of sources of EM radiation or RF interference such as Fluorescent Tube lights, speakers or steel support beams, below wall AC units which may drop water, next to louvre windows or where directly exposed to any source of UV light.
02. All works to be undertaken in line with the 'NDP Notes and Procedures' document and the Notes in this NDP.
03. The cabling installer is to mount and patch all WAPs, controllers (if required) and peripherals (PoE Injectors, PoE Midspans, Power rails, shelves). The installer may mount the wireless controller, power rail, or PoE injector(s) in the rear of the cabinet when there is not enough room in the front.
04. All WAP TOs are to be tested prior to WAP mounting to ensure TOs are suitable for installation. Refer to the TO testing procedure provided in the 'NDP Notes and Procedures' Document.
05. The installer is to re-shuffle the existing equipment to allow adequate room to mount the new switches.
06. The installer shall carefully dismount and remove all existing WAPs. The installer should not remove any existing trunking or neat cap used to run patch leads to the existing WAP unless the trunking pathway to be used is the same as the pre-existing pathway.

Site Wide Notes:

01. Buildings covered in blue shading are being remodelled under an external CSR project (Package A). Estimated construction period between November 2016 – September 2017. Once the new build is complete, the school can then redeploy The WAPs into the new spaces.
02. The CSR project also consists of a Package B Works which will involve replacement and construction of a new PAC and Science Hub. Estimated construction period between April 2017 – February 2019. Once the new build is complete, the school can then redeploy The WAPs into the new spaces.

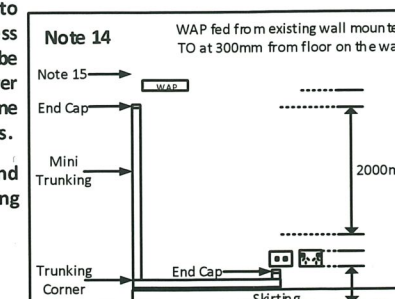


Overall Site Plan:



Installation Notes

07. WAPs mounted in this location to be protected against potential damage from physical education activities which may take place. Our recommended solution for this is an IP56 rated, perspex faced plastic enclosure and it is compulsory that the installer shall quote for this during as part of their RFP response and be installed during the project. Please refer to Sheet 3 (Examples 1c and 1d) for acceptable and unacceptable protection measures.
08. Exact TO locations are unknown in this location. The installer is review TO locations against the WAP Mounting Procedure supplied and select the most suitable location to mount the WAP.
09. The classroom indicated is outside the scope of SNUP due to current entitlement.
10. The installer shall allow to complete testing and mounting of existing WAPs in accordance with the methodology prescribed in the current 'NDP Notes and Installation Procedures' document for WAP mounting and patching.
11. The existing WAP indicated is outside the scope of this project and shall be retained as is.
12. The installer is required to ensure all WAPs are patched in to the switches in accordance with the method described on the page titled "Wireless Hardware Installation Notes"
13. The cabinet indicated has insufficient room to mount a cantilever shelf to support any PoE injectors for the wireless, the installer shall mount the injectors at the rear of the cabinet instead.
14. The installer shall install trunking as indicated for all WAPs fed from Low Level (LL) T/Os.
15. The installer shall coil any excess patch lead on top of the WAP bracket and secure neatly for ALL new WAPs installed.
16. These wireless access points are outside the scope of SNUP due to entitlement. If these wireless access points are required, they are to be 100% funded by the school to cover the supply, installation and integration of these wireless access points.
17. The contractor shall directly connect the WAP to the Cabinet at this location.
18. The building indicated in dashed outline is outside the scope of the SNUP as the Ministry does not hold majority stake ownership (0% PMG).
19. These wireless access points are outside the scope of SNUP due to entitlement. If these wireless access points are required, they are to be 100% funded by the school to cover the license extension and rebase line existing Aerohive Support for 5 years.
20. The integrator shall allow to extend licenses and rebaseline existing Aerohive Support for 5 years.



Legend

Total Quantity		
Symbol	Count	Description
	9	Retained "ac" WAP Aerohive AP230 (Notes 10 & 12 & 20)
	7	Existing WAP to be removed
	15	High Level Outlet
	8	New WAP (Notes 10 & 12)
	1	New "ac" WAP 100% school funded (Notes 10,12 & 16)
	1	Relocate existing WAP to new location indicated by arrow end - Note 08
	1	Retained "ac" WAP Aerohive AP230 Out of Scope (Notes 11 & 19)
	1	Low Level Outlet

Key

	Cabinet – Patch Panel location
CM	Ceiling Mount

Location: [0340] Cashmere High School

Contact: Carla Smith

Title:
ICT Infrastructure Wireless Access Point Locations

Designer: MaryAnne Malaluan

SIZE	Release Date
A3	12 April 2017
SCALE	
N/A	
DWG NO	
CHS-0340-1606-2450-02-4	
Job No. 1606-2450	REV
SHEET	4 of 10

Document History				
Drn.	Review	Date	Reason	Rev.
MAM	RS	4/4	Initial Draft	0.1

File Name: [School Initials-Ministry of Education #Job #, Rev]

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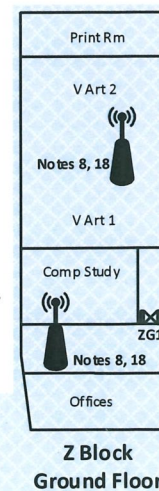
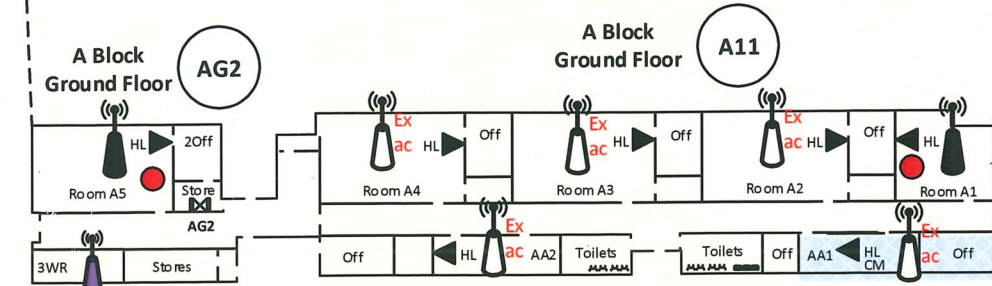
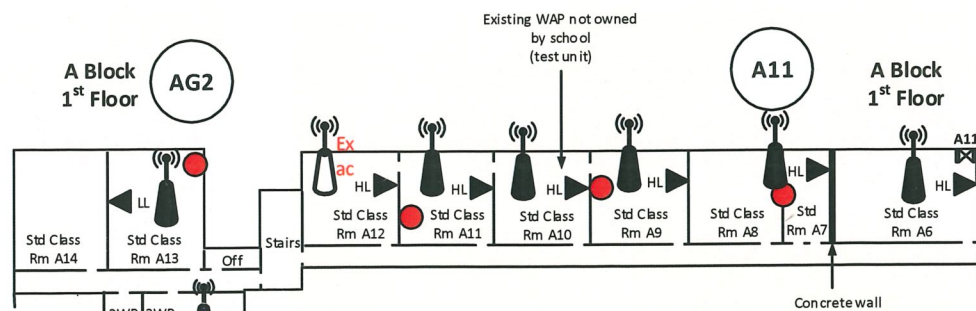
WIRELESS ACCESS POINT (WAP) REQUIREMENTS

General Notes (All Notes Apply)

- The plans below show the proposed location of WAPs. Do not mount WAPs back to back in adjacent classrooms, within 300mm of sources of EM radiation or RF interference such as Fluorescent Tube lights, speakers or steel support beams, below wall AC units which may drop water, next to louvre windows or where directly exposed to any source of UV light.
- All works to be undertaken in line with the 'NDP Notes and Procedures' document and the Notes in this NDP.
- The cabling installer is to mount and patch all WAPs, controllers (if required) and peripherals (PoE Injectors, PoE Midspans, Power rails, shelves). The installer may mount the wireless controller, power rails or PoE injector(s) in the rear of the cabinet when there is not enough room in the front.
- All WAP TOs are to be tested prior to WAP mounting to ensure TOs are suitable for installation. Refer to the TO testing procedure provided in the 'NDP Notes and Procedures' Document.
- The installer is to re-shuffle the existing equipment to allow adequate room to mount the new switches.
- The installer shall carefully dismantle and remove all existing WAPs. The installer should not remove any existing trunking or neat cap used to run patch leads to the existing WAP unless the trunking pathway to be used is the same as the pre-existing pathway.

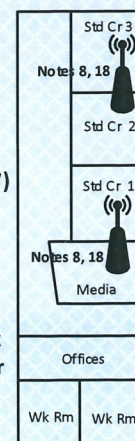
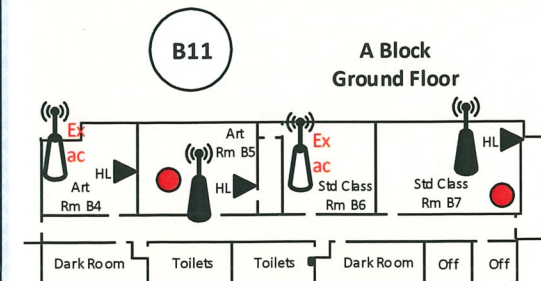
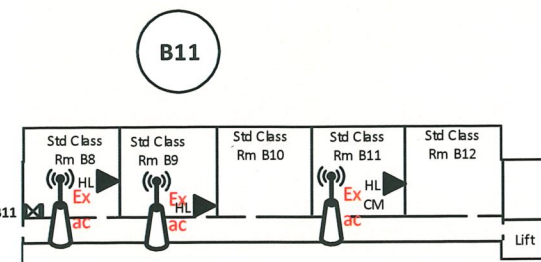
Site Wide Notes:

- Buildings covered in blue shading are being remodelled/added under an external CSR project (Package A). Estimated construction period between November 2016 – September 2017. Once the new build is complete, the school can then redeploy The WAPs into the new spaces.
- The CSR project also consists of a Package B Works which will involve replacement and construction of a new PAC and Science Hub. Estimated construction period between April 2017 – February 2019. Once the new build is complete, the school can then redeploy The WAPs into the new spaces.

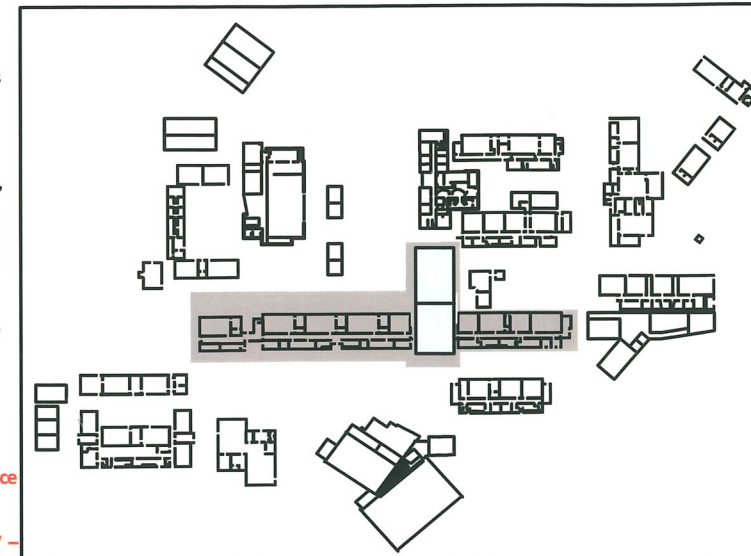


New Visual Arts and English Block Under Construction (completion May 2017)
Please refer to Site Wide Note 01

ZG1



Overall Site Plan:



Legend

Total Quantity		
Symbol	Count	Description
	8	Existing WAP to be removed
	11	Retained "ac" WAP Aerohive AP230 (Notes 10 & 12 & 20)
	1	Low Level Outlet
	14	New WAP (Notes 10 & 12)
	19	High Level Outlet
	2	Retained "ac" WAP Aerohive AP230 Out of Scope (Notes 11 & 19)

Installation Notes

- WAPs mounted in this location to be protected against potential damage from physical education activities which may take place. Our recommended solution for this is an IP56 rated, perspex faced plastic enclosure and it is compulsory that the installer shall quote for this during as part of their RFP response and be installed during the project. Please refer to Sheet 3 (Examples 1c and 1d) for acceptable and unacceptable protection measures.
- Exact TO locations are unknown in this location. The installer is review TO locations against the WAP Mounting Procedure supplied and select the most suitable location to mount the WAP.
- The classroom indicated is outside the scope of SNUP due to current entitlement.
- The installer shall allow to complete testing and mounting of existing WAPs in accordance with the methodology prescribed in the current 'NDP Notes and Installation Procedures' document for WAP mounting and patching.
- The existing WAP indicated is outside the scope of this project and shall be retained as is.
- The installer is required to ensure all WAPs are patched in to the switches in accordance with the method described on the page titled "Wireless Hardware Installation Notes"
- The cabinet indicated has insufficient room to mount a cantilever shelf to support any PoE injectors for the wireless, the installer shall mount the injectors at the rear of the cabinet instead.
- The installer shall install trunking as indicated for all WAPs fed from Low Level (LL) T/Os.
- The installer shall coil any excess patch lead on top of the WAP bracket and secure neatly for ALL new WAPs installed.
- These wireless access points are outside the scope of SNUP due to entitlement. If these wireless access points are required, they are to be 100% funded by the school to cover the supply, installation and integration of these wireless access points.
- The contractor shall directly connect the WAP to the Cabinet at this location.

- The block indicated in blue shading is under construction due for completion May 2017. Should the new building space be accessible at the time of the WSUP project mobilizing at the school, the installer shall undertake the upgrade as per standard WSUP procedure. Alternatively the 3 x WAPs (and 1 x out of scope WAP) allocated to these locations shall be temporarily patched to the edge switch in cabinet B11 for the purposes of WLAN commissioning. Once remote configuration by the integrator is complete, the installer shall unpatch, place in packaging and return to the school for safekeeping. Once the new build is complete, the school can then redeploy the hardware into the new spaces.
- The integrator shall allow to extend licenses and rebaseline existing Aerohive Support for 5 years.

Key	
	Cabinet – Patch Panel location
CM	Ceiling Mount

Location: [0340] Cashmere High School

Contact: Carla Smith

Title:
ICT Infrastructure Wireless Access Point Locations

Designer: MaryAnne Malaluan

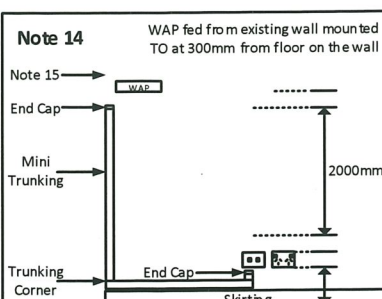
SIZE	Release Date
A3	12 April 2017
SCALE	N/A
DWG NO	REV
CHS-0340-1606-2450-02-5	0.2
Job No. 1606-2450	SHEET 5 of 10

Document History				
Drn.	Review	Date	Reason	Rev.
MAM	RS	4/4	Initial Draft	0.1

File Name: [School Initials-Ministry of Education #Job #, Rev]

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WIRELESS ACCESS POINT (WAP) REQUIREMENTS

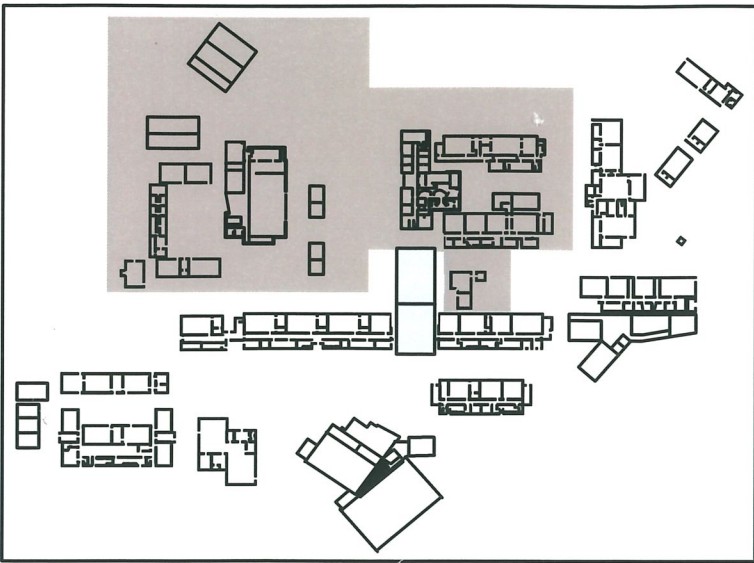
General Notes (All Notes Apply)

01. The plans below show the proposed location of WAPs. Do not mount WAPs back to back in adjacent classrooms, within 300mm of sources of EM radiation or RF interference such as Fluorescent Tube lights, speakers or steel support beams, below wall AC units which may drop water, next to louvre windows or where directly exposed to any source of UV light.
02. All works to be undertaken in line with the 'NDP Notes and Procedures' document and the Notes in this NDP.
03. The cabling installer is to mount and patch all WAPs, controllers (if required) and peripherals (PoE Injectors, PoE Midspans, Power rails, shelves). The installer may mount the wireless controller, power rails or PoE injector(s) in the rear of the cabinet when there is not enough room in the front.
04. All WAP TOs are to be tested prior to WAP mounting to ensure TOs are suitable for installation. Refer to the TO testing procedure provided in the 'NDP Notes and Procedures' Document.
05. The installer is to re-shuffle the existing equipment to allow adequate room to mount the new switches.
06. The installer shall carefully dismount and remove all existing WAPs. The installer should not remove any existing trunking or neat cap used to run patch leads to the existing WAP unless the trunking pathway to be used is the same as the pre-existing pathway.

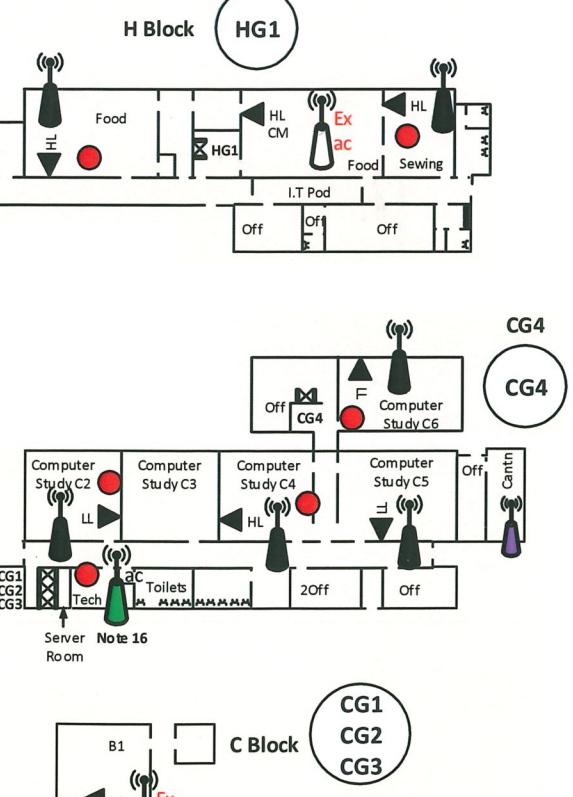
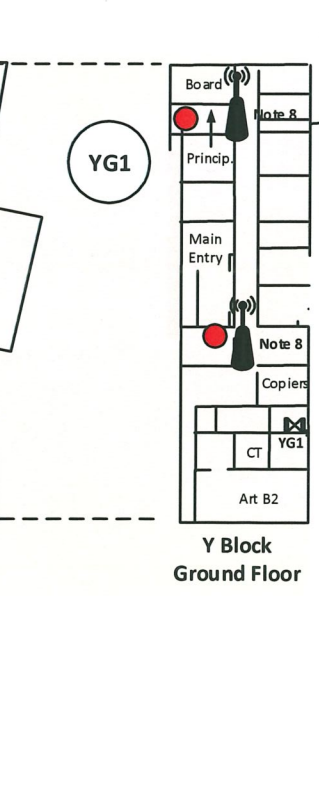
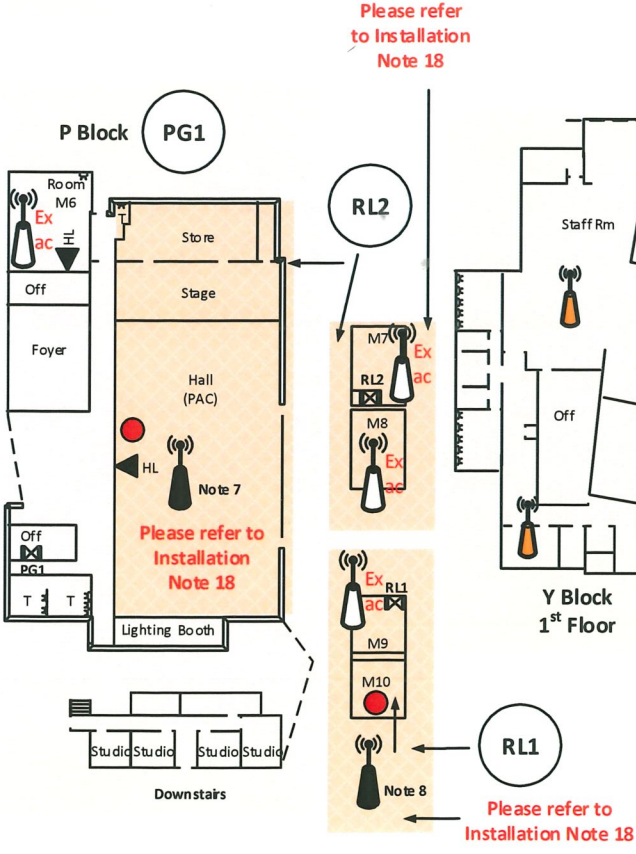
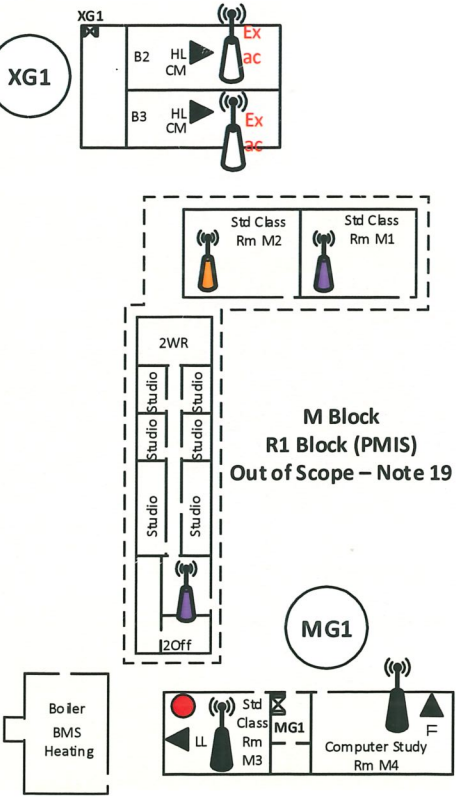
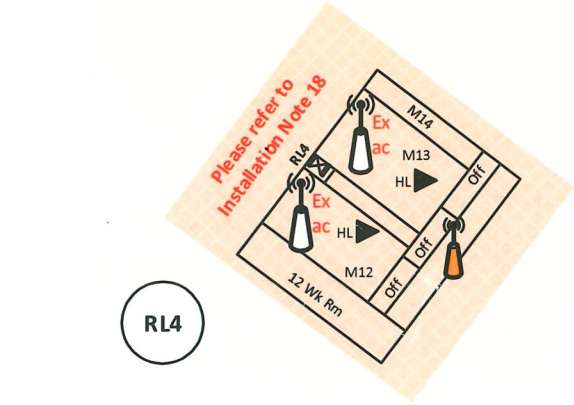
Installation Notes

07. WAPs mounted in this location to be protected against potential damage from physical education activities which may take place. Our recommended solution for this is an IP56 rated, perspex faced plastic enclosure and it is compulsory that the installer shall quote for this during as part of their RFP response and be installed during the project. Please refer to Sheet 3 (Examples 1c and 1d) for acceptable and unacceptable protection measures.
08. Exact TO locations are unknown in this location. The installer is review TO locations against the WAP Mounting Procedure supplied and select the most suitable location to mount the WAP.
09. The classroom indicated is outside the scope of SNUP due to current entitlement.
10. The installer shall allow to complete testing and mounting of existing WAPs in accordance with the methodology prescribed in the current 'NDP Notes and Installation Procedures' document for WAP mounting and patching.
11. The existing WAP indicated is outside the scope of this project and shall be retained as is.

Overall Site Plan:



Legend		
Total Quantity		
Symbol	Count	Description
	11	Existing WAP to be removed
	10	Retained "ac" WAP Aerohive AP230 (Notes 10 & 12 & 21)
	11	High Level Outlet
	12	New WAP (Notes 10 & 12)
	5	Low Level Outlet
	1	New "ac" WAP 100% school funded (Notes 10,12 & 16)
	4	Retained "n" WAP Aerohive AP330 Out of Scope (Notes 11 & 20)
	3	Retained "ac" WAP Aerohive AP230 Out of Scope (Notes 11 & 20)



12. The installer is required to ensure all WAPs are patched in to the switches in accordance with the method described on the page titled "Wireless Hardware Installation Notes"
13. The cabinet indicated has insufficient room to mount a cantilever shelf to support any PoE injectors for the wireless, the installer shall mount the injectors at the rear of the cabinet instead.
14. The installer shall install trunking as indicated for all WAPs fed from Low Level (LL) T/Os.
15. The installer shall coil any excess patch lead on top of the WAP bracket and secure neatly for ALL new WAPs installed.
16. These wireless access points are outside the scope of SNUP due to entitlement. If these wireless access points are required, they are to be 100% funded by the school to cover the supply, installation and integration of these wireless access points.
17. The contractor shall directly connect the WAP to the Cabinet at this location.

Key	
	Cabinet - Patch Panel location
CM	Ceiling Mount

Location: [0340] Cashmere High School

Contact: Carla Smith

Title: ICT Infrastructure Wireless Access Point Locations

Designer: MaryAnne Malaluan

SIZE	Release Date
A3	12 April 2017
SCALE	N/A
DWG NO	REV
CHS-0340-1606-2450-02-6	0.2
Job No. 1606-2450	SHEET 6 of 10

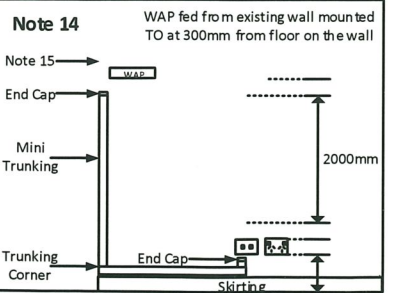
Document History				
Drn.	Review	Date	Reason	Rev.
MAM	RS	4/4	Initial Draft	0.1

File Name: [School] Initials-Ministry of Education #Job #, Rev]

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18. The blocks indicated in orange shading are temporary and are to be removed as part of a CSR Project (dates TBC). Should the Removals be active at the time of the WSNUP project mobilizing at the school, the installer shall temporarily patch the WAPs to the edge switch in the cabinets for the purposes of WLAN commissioning. Once remote configuration by the integrator is complete, the installer shall un-patch, place in packaging and return to the school for safekeeping. Once the new build is complete, the school can then redeploy the hardware into the new spaces.
19. The building indicated in dashed outline is outside the scope of the SNUP as the Ministry does not hold majority stake ownership (0% PMG).
20. These wireless access points are outside the scope of SNUP due to entitlement. If these wireless access points are required, they are to be 100% funded by the school to cover the license extension and rebase line existing Aerohive Support for 5 years.
21. The integrator shall allow to extend licenses and rebase line existing Aerohive Support for 5 years.



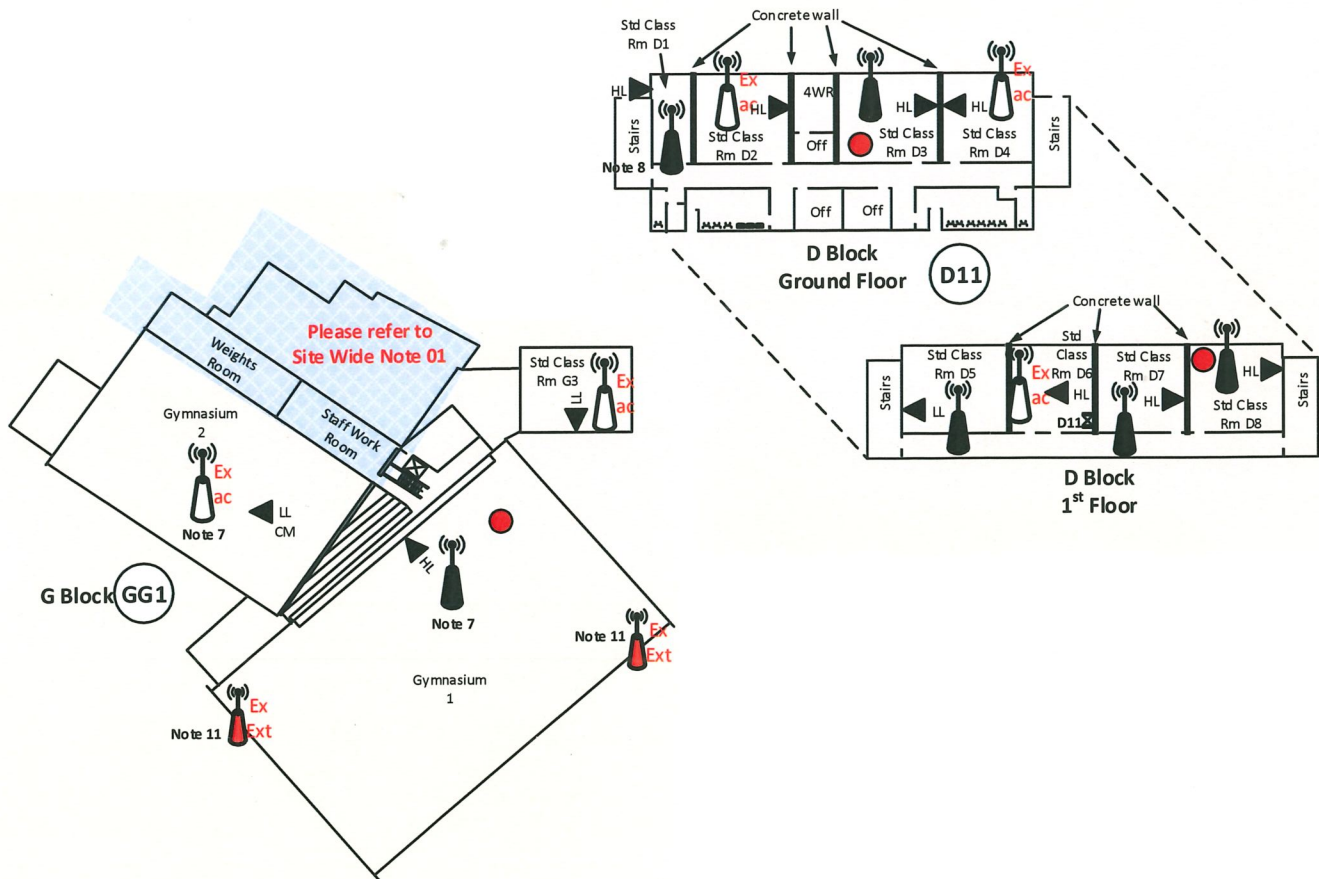
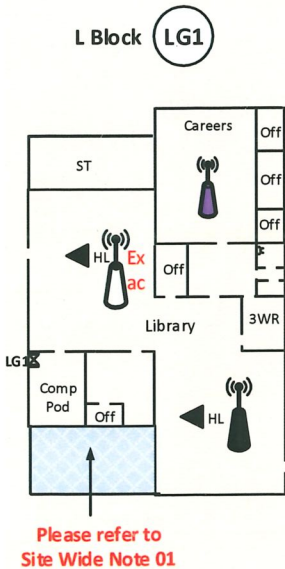
WIRELESS ACCESS POINT (WAP) REQUIREMENTS

General Notes (All Notes Apply)

01. The plans below show the proposed location of WAPs. Do not mount WAPs back to back in adjacent classrooms, within 300mm of sources of EM radiation or RF interference such as Fluorescent Tube lights, speakers or steel support beams, below wall AC units which may drop water, next to louvre windows or where directly exposed to any source of UV light.
02. All works to be undertaken in line with the 'NDP Notes and Procedures' document and the Notes in this NDP.
03. The cabling installer is to mount and patch all WAPs, controllers (if required) and peripherals (PoE Injectors, PoE Midspans, Power rails, shelves). The installer may mount the wireless controller, power rails or PoE injector(s) in the rear of the cabinet when there is not enough room in the front.
04. All WAP TOs are to be tested prior to WAP mounting to ensure TOs are suitable for installation. Refer to the TO testing procedure provided in the 'NDP Notes and Procedures' Document.
05. The installer is to re-shuffle the existing equipment to allow adequate room to mount the new switches.
06. The installer shall carefully dismount and remove all existing WAPs. The installer should not remove any existing trunking or neat cap used to run patch leads to the existing WAP unless the trunking pathway to be used is the same as the pre-existing pathway.

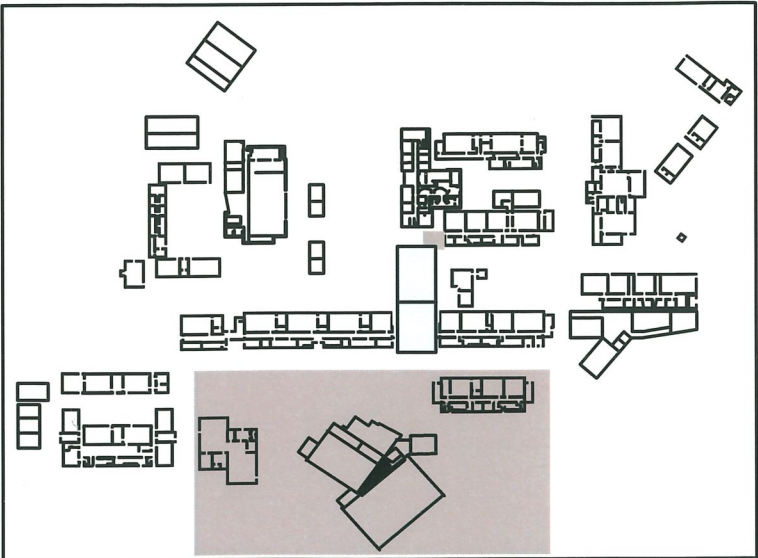
Site Wide Notes:

01. Buildings covered in blue shading are being remodelled/added under an external CSR project (Package A). Estimated construction period between November 2016 – September 2017. Once the new build is complete, the school can then redeploy The WAPs into the new spaces.
02. The CSR project also consists of a Package B Works which will involve replacement and construction of a new PAC and Science Hub. Estimated construction period between April 2017 – February 2019. Once the new build is complete, the school can then redeploy The WAPs into the new spaces.



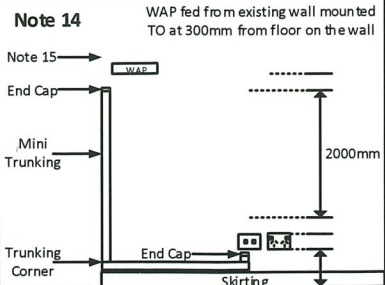
18. The integrator shall allow to extend licenses and rebase line existing Aerohive Support for 5 years.
19. These wireless access points are outside the scope of SNUP due to entitlement. If these wireless access points are required, they are to be 100% funded by the school to cover the license extension and rebase line existing Aerohive Support for 5 years.
20. The integrator shall allow to extend licenses and rebase line existing Aerohive Support for 5 years.

Overall Site Plan:



Installation Notes

07. WAPs mounted in this location to be protected against potential damage from physical education activities which may take place. Our recommended solution for this is an IP56 rated, perspex faced plastic enclosure and it is compulsory that the installer shall quote for this during as part of their RFP response and be installed during the project. Please refer to Sheet 3 (Examples 1c and 1d) for acceptable and unacceptable protection measures.
08. Exact TO locations are unknown in this location. The installer is review TO locations against the WAP Mounting Procedure supplied and select the most suitable location to mount the WAP.
09. The classroom indicated is outside the scope of SNUP due to current entitlement.
10. The installer shall allow to complete testing and mounting of existing WAPs in accordance with the methodology prescribed in the current 'NDP Notes and Installation Procedures' document for WAP mounting and patching.
11. The existing WAP indicated is outside the scope of this project and shall be retained as is.
12. The installer is required to ensure all WAPs are patched in to the switches in accordance with the method described on the page titled "Wireless Hardware Installation Notes"
13. The cabinet indicated has insufficient room to mount a cantilever shelf to support any PoE injectors for the wireless the installer shall mount the injectors at the rear of the cabinet instead.
14. The installer shall install trunking as indicated for all WAPs fed from Low Level (LL) T/Os.
15. The installer shall coil any excess patch lead on top of the WAP bracket and secure neatly for ALL new WAPs installed.
16. These wireless access points are outside the scope of SNUP due to entitlement. If these wireless access points are required, they are to be 100% funded by the school to cover the supply, installation and integration of these wireless access points.
17. The contractor shall directly connect the WAP to the Cabinet at this location.



Legend		
Total Quantity		
Symbol	Count	Description
	6	Retained "ac" WAP Aerohive AP230 (Notes 10 & 12 & 20)
	3	Existing WAP to be removed
	2	Retained external WAP (Note 11)
	7	New WAP (Notes 10 & 12)
	10	High Level Outlet
	3	Low Level Outlet
	1	Retained "ac" WAP Aerohive AP230 Out of Scope (Notes 11 & 19)

Key	
	Cabinet – Patch Panel location
CM	Ceiling Mount

Location: [0340] Cashmere High School

Contact: Carla Smith

Title:
ICT Infrastructure Wireless Access Point Locations

Designer: MaryAnne Malaluan

SIZE	Release Date
A3	12 April 2017

SCALE	DWG NO	REV
N/A	CHS-0340-1606-2450-02-7	0.2
Job No. 1606-2450	SHEET	7 of 10

Document History				
Dm.	Review	Date	Reason	Rev.
MAM	RS	4/4	Initial Draft	0.1

File Name: [School Initials-Ministry of Education #Job #, Rev]

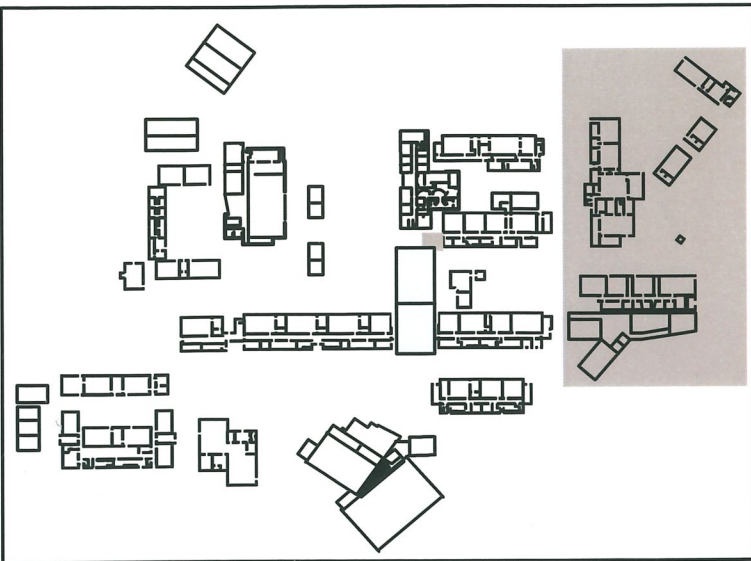
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WIRELESS ACCESS POINT (WAP) REQUIREMENTS

General Notes (All Notes Apply)

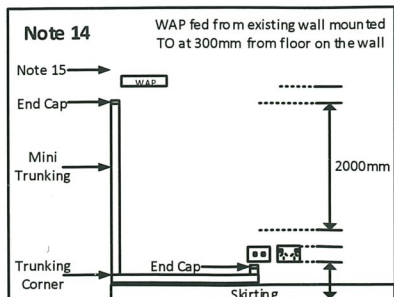
01. The plans below show the proposed location of WAPs. **Do not mount WAPs back to back in adjacent classrooms, within 300mm of sources of EM radiation or RF interference such as Fluorescent Tube lights, speakers or steel support beams, below wall AC units which may drop water, next to louvre windows or where directly exposed to any source of UV light.**
02. All works to be undertaken in line with the 'NDP Notes and Procedures' document and the Notes in this NDP.
03. The cabling installer is to mount and patch all WAPs, controllers (if required) and peripherals (PoE Injectors, PoE Midspans, Power rails, shelves). **The installer may mount the wireless controller, power rails or PoE injector(s) in the rear of the cabinet when there is not enough room in the front.**
04. All WAP TOs are to be tested prior to WAP mounting to ensure TOs are suitable for installation. Refer to the TO testing procedure provided in the 'NDP Notes and Procedures' Document.
05. The installer is to re-shuffle the existing equipment to allow adequate room to mount the new switches.
06. **The installer shall carefully dismount and remove all existing WAPs. The installer should not remove any existing trunking or neat cap used to run patch leads to the existing WAP unless the trunking pathway to be used is the same as the pre-existing pathway.**

Overall Site Plan:



Installation Notes

07. WAPs mounted in this location to be protected against potential damage from physical education activities which may take place. Our recommended solution for this is an IP56 rated, perspex faced plastic enclosure and it is compulsory that the installer shall quote for this during as part of their RFP response and be installed during the project. Please refer to Sheet 3 (Examples 1c and 1d) for acceptable and unacceptable protection measures.
08. Exact TO locations are unknown in this location. The installer is review TO locations against the WAP Mounting Procedure supplied and select the most suitable location to mount the WAP.
09. The classroom indicated is outside the scope of SNUP due to current entitlement.
10. The installer shall allow to complete testing and mounting of existing WAPs in accordance with the methodology prescribed in the current 'NDP Notes and Installation Procedures' document for WAP mounting and patching.
11. The existing WAP indicated is outside the scope of this project and shall be retained as is.
12. The installer is required to ensure all WAPs are patched in to the switches in accordance with the method described on the page titled "Wireless Hardware Installation Notes"
13. The cabinet indicated has insufficient room to mount a cantilever shelf to support any PoE injectors for the wireless, the installer shall mount the injectors at the rear of the cabinet instead.
14. The installer shall install trunking as indicated for all WAPs fed from Low Level (LL) T/Os.
15. The installer shall coil any excess patch lead on top of the WAP bracket and secure neatly for ALL new WAPs installed.
16. These wireless access points are outside the scope of SNUP due to entitlement. If these wireless access points are required, they are to be 100% funded by the school to cover the supply, installation and integration of these wireless access points.
17. The contractor shall directly connect the WAP to the Cabinet at this location.



Legend

Total Quantity		
Symbol	Count	Description
	7	Existing WAP to be removed
	6	Retained "ac" WAP Aerohive AP230 (Notes 10 & 12 & 20)
 LL	2	Low Level Outlet
	10	New WAP (Notes 10 & 12)
 HL	12	High Level Outlet
	1	Retained "ac" WAP Aerohive AP230 Out of Scope (Notes 11 & 19)

Key

	Cabinet – Patch Panel location
CM	Ceiling Mount

Location: [0340] Cashmere High School

Contact: Carla Smith

Title:
ICT Infrastructure Wireless Access Point Locations

Designer: MaryAnne Malaluan

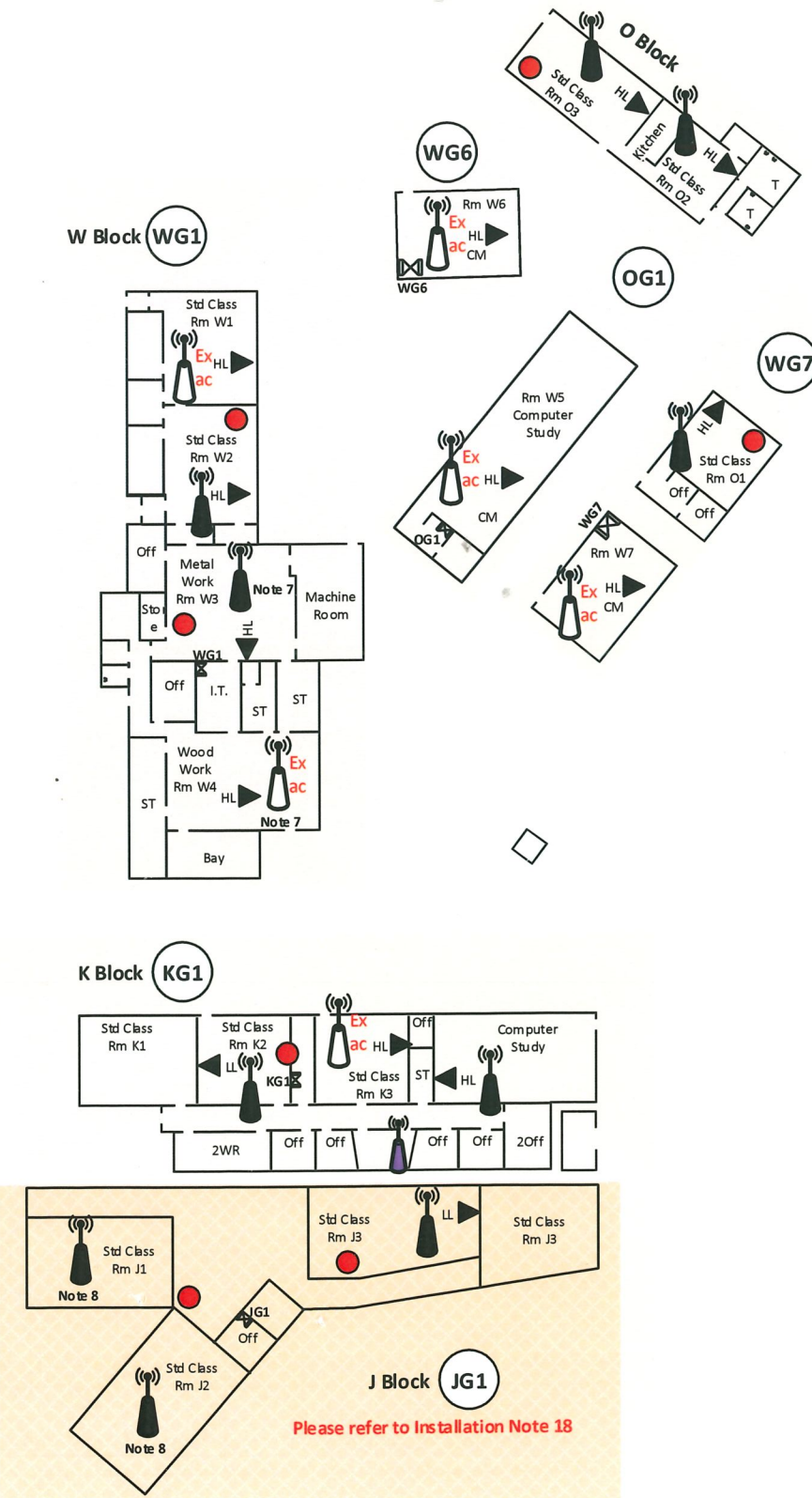
SIZE A3	Release Date 12 April 2017		
SCALE		N/A	
DWG NO CHS-0340-1606-2450-02-8			REV 0.2
Job No. 1606-2450		SHEET	8 of 10

Document History

[illegible]

File Name: [School Initials-Ministry of Education #-Job #, Rev]

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18. The blocks indicated in orange shading Are temporary and are to be removed as part of a CSR Project (dates TBC). Should the Removals be active at the time of the WSNUP project mobilizing at the school, the installer shall temporarily patch the WAPs to the edge switch in the cabinets for the purposes of WLAN commissioning. Once remote configuration by the integrator is complete, the installer shall unpatch, place in packaging and return to the school for safekeeping. Once the new build is complete, the school can then redeploy the hardware into the new spaces.

19. These wireless access points are outside the scope of SNUP due to entitlement. If these wireless access points are required, they are to be 100% funded by the school to cover the license extension and rebase line existing Aerohive Support for 5 years.

20. The integrator shall allow to extend licenses and rebaseline existing Aerohive Support for 5 years.

WAP MOUNTING AND PATCHING PROCEDURE

1. All WAPs, brackets, and patch cords are to be supplied by others.
2. **The installer shall carefully dismount and remove all existing WAPs. The installer should not remove any existing trunking or neat cap used to run patch leads to the existing WAP unless the trunking pathway to be used as part of this project will be the same as the pre-existing pathway. Place all old WAP hardware in the Main Cabinet Room in a tidy manner.**
3. If the school's wireless solution utilises an existing controller, the installer shall power down and disconnect the controller from the network. The old controller should be removed from the cabinet and stacked neatly with the WAPs.
4. Where school existing WAPs are being retained in this project, the installer shall consider it as if it was a new WAP to be installed. The WAP should be unmounted and disconnected from its T/O, the T/O should be tested, and the WAP remount as per the procedures detailed below.
5. The installer shall discuss all trunking pathways with the school's authorised point of contact before any pathways are installed for the new WAPs. Where possible the installer should install WAPs at the locations shown on the NDP, however, the school's preferred WAP location within the same room is also acceptable. It is the responsibility of the installer to assess the proposed location and confirm it suitable to mount the WAP.
6. The installer shall ensure that the trunking pathway chosen shall not unnecessarily bisect, walls, CRM display/feature walls, pin boards and follow the natural lines of the wall where possible. Refer to the illustration below for an acceptable (Example 1a) and incorrect install (Example 1b).
7. When deciding on the best T/O to utilise in a classroom or school facility; please follow the following order of preference:
 - a. High level whiteboard T/O
 - b. Ceiling mounted T/O
 - c. Low level T/O – as indicated on the NDP
 - d. Low level T/O – the next closest to the one indicated on the NDP.

In the event of a failed T/O which cannot be rectified, the installer is required to utilise the next preferred outlet.

8. The following items should be considered when reviewing WAP mounting locations:
 - a. Do not mount WAPs back to back in adjacent rooms, within 300mm of EM radiation or RF interference such as Fluorescent Tube lights, speakers or steel support beams.
 - b. Do not mount WAPs in locations where they may be prone to damage i.e. below wall AC units which may drop water, next to louvre windows or where directly exposed to any source of UV light.
 - c. WAPs should preferably be mounted on internal wall as close as possible to the centre of the room.
 - d. If high level outlets are available these should take precedence over any low level TOs assuming none of the above is an issue.
9. Once a suitable location has been selected the installer should test the chosen TO as per the "TELECOMMUNICATIONS OUTLETS (TO) – TESTING & REPAIR PROCEDURE".
10. If the chosen outlet is in use by another device then this device should be relocated to another location in the room.
11. The distance between the WAP and T/O should be in general less than 3 metres but should not exceed 5 metres.
12. The WAP should be mount a minimum of 2.2m above floor level directly above the TO. The WAP should not be mounted closer than 300mm to the ceiling.
13. The installer shall mount the WAP horizontally off the wall using the supplied 'L bracket'. Specific mounting instructions will be provided to the installer by the WLAN integrator as part of the wireless hardware delivered.
14. A length of small form factor (40mmx25mm) trunking shall be installed to migrate the patch cord from the low level T/O to the WAP. Appropriate manufacture end caps shall be installed on both ends of the trunking with a small hole installed to allow for the patch cord entry and exit. The use of neat cap is not permitted.
15. A 3 metre (or 5 metre) White Cat 6 patch cord shall be neatly installed between the T/O and WAP and run within the installed trunking of point 13 above. Any excess cord length should be hidden inside the trunking.
16. The installer shall record the necessary details in the Hardware Installation Schedule for the newly installed WAP.
17. Where WAPs are installed inside Physical Education areas the installer shall mount the WAP inside a protective plastic enclosure.

Acceptable and Incorrect Trunking Pathway Examples



Example 1a: acceptable trunking pathway installed to follow edge of pin board.



Example 1b: Incorrect trunking pathway installed bisecting pin board.

18. When an access point is mounted near a HL (High level TO) it shall be within 0.5m of the TO due to the standard patch cord length supplied and with a LL (Low level TO or Note 8) the WAP shall be mounted within a 4m trunking run of the TO as 5m patch cords are supplied standard.

PERIPHERALS/CONTROLLER MOUNTING AND PATCHING PROCEDURE

1. All controllers, shelves, power injectors, mid-span injectors, power rails and patch cords shall be supplied by others.
2. Refer to the "Hardware Allocation Table" for a detailed breakdown of all hardware on a per cabinet basis.
3. **If the school's wireless solution utilises an existing controller, the installer shall power down and disconnect the controller from the network. The old controller should be removed from the cabinet and stacked neatly with the WAPs.**
4. If a controller is provided as part of the WLAN hardware the installer shall mount this into the school server cabinet in the next available Rack Unit (RU). The controller uplink port shall be patched directly into a copper port on the core switch using a white patch cord.
5. Once the controller has been patched into the switch it should then be plugged in and powered up (via a power rail or cabinet GPO).
6. At each distribution cabinet the installer shall tidily mount power injectors in place. Where there are three or more power injectors these shall be mounted onto a cantilever shelf and secured in place with Velcro ties. The cantilever shelf shall be installed in the next available rack unit (RU) inside the cabinet. Where there are two or less power injectors these shall be either left in the base of the cabinet or secured to the rear mounting rails.
7. In cabinets servicing six or more WAPs a mid-span injector will be provided to replace the power injectors and cantilever shelf. The mid-span injector can support up to 12 WAPs.
8. White patch cords shall be used to patch between the power Injectors/mid-span and the TO and network switch.
9. Where appropriate a six way power rail will be provided and the installer is to mount this in the rear of the cabinet to power the injectors, mid-spans, network switches and controllers (if installed). Power cabling for all installed equipment shall be tidily managed in the rear of the cabinet and adequate clearance shall be maintained from data cabling.
10. The installer shall note all required patching information inside the "Hardware Installation Schedule".
11. Once all WAPs have been powered up and any controllers installed, the installer shall contact the WLAN integrator and confirm all WAPs are visible on the network. In the event there are missing WAPs the installer shall double check all WAPs are powered up and patched into the switch. N + 1 power injectors shall be provided in the event of a power injector failure. Any faulty power injectors shall be returned to Allied Telesis follow the installation. The installer is not to remove the spare injector from site – this is the school's property.

12. Before seeking sign off the installer shall confirm the school can access the new WLAN via a school device.
13. All old WAPs that do not form part of the new solution are to be removed by the WSNUP Installer once wireless implementation has been completed.

TELECOMMUNICATIONS OUTLETS (TO) – TESTING & REPAIR PROCEDURE

1. No new outlets are to be installed during this installation process.
2. Only T/Os proposed for WAP use shall be tested. All test results shall be provided to Torque IP on completion of the installation.
3. Test top data outlet of the chosen dual outlet indicated on WAP layout.
4. If outlet passes to appropriate standard (Class D minimum), then proceed to mount WAP and patch as per "WAP MOUNTING AND PATCHING PROCEDURE".
5. If outlet fails or provides marginal result, test bottom outlet. If outlet passes, use for WAP patching as per "WAP MOUNTING AND PATCHING PROCEDURE"
6. If this bottom outlet fails as well, then assess the test results for the fault source. If repair is possible at either end via re-termination, then proceed to repair. Test results must show the before and after results for this outlet, as any repairs shall be processed as variations.
7. If both outlets fail and cannot be repaired via re-termination of the TO or patch panel then the installer shall review the WAP layout and move to the next location preference.

Acceptable and unacceptable Hall/Gym WAP protection



Example 1c: Acceptable WAP protection installed with IP56 rated perspex faced plastic enclosure.



Example 1d: Unacceptable WAP protection installed with metal caging and lack of aesthetics.

Wireless Hardware Installation & Commissioning Notes

Contact: Carla Smith

Designer: MaryAnne Malaluan			
SIZE A3	Release Date 12 April 2017		
SCALE	N/A		
DWG NO CHS-0340-1606-2450-02-3			REV 0.2
Job No. 1606-2450		SHEET	3 of 10

GENERAL NOTES

1.1 WIRELESS INSTALLATION METHODOLOGY

- The installer is required to obtain confirmation from Allied Telesis & the Wireless Integrator before leaving site that the switching and wireless networks are operational.
- The order of site works is critical to ensure both onsite and remote parties have sufficient time to complete their work. Site works shall be completed in the following order:
 - On the morning of the installation, The lead installer must spend 15 – 20 minutes with the School's point of contact and cover all points on the Preinstall Checklist.
 - The installer shall provide the school contact with details of the school's expected network outage.
 - The installer shall ensure that photos of all cabinets (front and back) are taken prior to commencing any WSNUP installation work. This is a mandatory requirement and any issues that arise due to faults reported post upgrade (i.e. incorrect patching) may require return visit by the installer to resolve at their own cost.
 - Power down the school router and then Complete "SWITCH MOUNTING AND PATCHING PROCEDURE" for all cabinets.
 - Power up school router.
 - Complete "SWITCH COMMISSIONING PROCEDURE".
 - The switch commissioner will advise the installer once the switch network is fully operational.
 - While the switch commissioner is completing remote testing, the installer shall complete "WAP MOUNTING AND PATCH PROCEDURE" for all WAPs.
 - Discuss WAP location and trunking pathway with School point of contact.
 - Test T/Os before beginning installation
 - Install trunking as per the pathway discussed with School.
 - Mount L Bracket and WAPs
 - Complete "PERIPHERALS AND CONTROLLER MOUNTING AND PATCHING PROCEDURE".
 - Contact the WLAN Integrator with remote access and confirm all WAPs and Controllers are visible on the school network.
 - Complete Installation Sign off with the school and ensure all documentation is in order before leaving site.

1.2 WSNUP GENERAL REQUIREMENTS AND INFORMATION

- All installation work shall comply with the current Ministry of Education ICT Infrastructure Standards.
- The Installer Contract pricing is final, and no project variations will be considered, for aspects missed, or not anticipated. The only agreed potential variations are for the elements identified in the rate card and tender documents.
- The Installer is to control all debris and waste products from the installation process, and is to remove these from site at the end of each day. Drop cloths, warning tapes, markers and cones are to be used where appropriate. The Installer is to provide all of his own cleaning equipment, and is not to utilise the schools. The Installer is liable for all restoration work (this however is expected to be nil if installed correctly).
- It is the responsibility of the Installer to mitigate discussions relating to structured cabling, backbones and pathways onsite. These are out of the scope of this wireless project and no such work shall be completed while the installer is onsite. If required the installer may schedule a revisit to complete any additional work requested by the school. Any additional work shall be completed at the school's cost.
- All centrally supplied hardware and wireless equipment shall be sent to the installer depot prior to the commencement of the project. It is the responsibility of the installer to confirm all hardware has been supplied prior to attending site. A "Goods Inwards Receipt" should be completed by the installer and uploaded to the Torque IP Portal as proof all equipment has been received by the installer for each school. Once the installer has taken possession of these goods they are then responsible for this hardware until the "Site Installation Sign-off" is completed by the school. The installer shall ensure they have adequate insurance in place to cover all equipment in the event of theft or damage.
- All site works shall be completed during standard business hours.
- A Hardware Reference Table is provided via the portal for each school with a detailed breakdown of all hardware on a per cabinet basis.
- Spare central procurement items will be supplied on a per school basis for large schools and a per cluster basis for small to medium schools. The spare items will include an assortment of copper and fibre patch cords as well as spare power rails. These spares should be used in the event the central procurement hardware allocated does not meet the specific requirements of the installation (i.e. supplied patch cords are incorrect lengths). For large schools all unused spares should be left onsite at the completion of the project. For clusters, the spares shall be taken by the installer and set aside for use in their next cluster. There is no requirement to return the spares to JA Russell.
- The installer shall ensure necessary time is included inside their project price (Rate Card – Base Price) to cover any troubleshooting of faulty backbone cabling. Repair of any damaged backbone cabling is out of scope of this project. In the event there is a potential fault causing a switch uplink not to function then the installer shall allow time to complete the necessary testing and diagnostics while onsite to confirm the cause of the fault. Such issues should be highlighted to the Torque IP Project Manager who will discuss a course of action with the appropriate party.

SWITCH MOUNTING (if applicable) AND PATCHING PROCEDURE

- Any switches and active hardware will be supplied by others.
- The installer must advise the school authorised contact of the approximate time and duration of the network downtime while any new switches (Where applicable) are installed.
- Installer shall take photos of all cabinets before commencing any work. The photos should show complete cabinet views as well as close up photos with the complete subject in frame. Close up photos shall show all patch panels, existing switches, and all other critical hardware (servers, routers, firewalls, media converters, old controllers, PoE Injectors, UPS units etc.) Installer should ensure that the fibre and copper leads can be traced in the event of troubleshooting the network after an upgrade.
- At the Main Cabinet (CD), the installer is to create a sketch of the current network as it is currently physically connected. All connections from the core switch to each backbone link (may be copper or fibre), to existing router(s), server(s), firewall(s), wireless controller and other critical network devices, should be noted.
- To begin any switch replacement, the installer should first power down and disconnect all school routers.
- The installer shall check the NDP Network Schematic for any existing switches that are to be retained as part of this project. In general, all retained switches should be retained as is and are excluded from the requirements of the following point 7. In some cases, the retained switch may need to be relocated to another cabinet.
- At each cabinet that the NDP Network Schematic requires a new switch to be install, the installer shall power down all existing switches, un-patch all patch cords from the switch ports and remove switches from the cabinet. All existing switches removed site wide shall be left in a tidy manner in the main cabinet room. The school authorised contact is to be notified of their location prior to leaving site. Note: Old existing switches must not be left connected to the network or in the cabinet under any circumstances. These older switches must be removed to ensure the quality of the network upon completion of this project.
- The new switches (Where applicable) shall then be mounted inside the cabinet directly replacing any removed switches. If a Power Rail (PDU) has been supplied for the cabinet, then the switch should be supplied from the new PDU. If no Power Rail supplied, then the switch shall be plugged directly into a GPO on the wall at the back of the cabinet.
- Where multiple switches are installed in any one cabinet (excluding x210 series), Care shall be taken when these switches are stacked together using the supplied stacking cables to avoid any network issues to incorrect stacking. Refer to the switch schematic and correct stacking methodology supplied by Allied Telesis for more information.
- In larger schools the installer may be required to mount and connect a redundant power supply to accompany the core switch. The installer shall follow all instructions supplied by Allied Telesis and if in doubt, shall contact Allied Telesis before proceeding.
- The installer shall then patch any switch uplinks as per the switch schematic supplied by Allied Telesis. Note: as part of this project, new fibre and copper patch cords have been supplied for new equipment only.
- At the Main Cabinet, the installer shall then reconnect all existing network hardware as it used to be and noted in point 4 above, any connections to the old switch (core or edge) should now be made to the new switch.
- At the Main Cabinet, all WAPs and Wireless controller (if applicable) shall be patched into the core switch.
- At Edge Cabinets, the WAP shall be patched to the top switch using the supplied white patch cords. Patching shall start from the very last copper port, and continue backwards sequentially until all WAPs are connected. Note: On some switches (i.e. x210 switches) the last ports may be shared with the SFP slots, if these slots are used for uplinks then the corresponding copper port will not be usable.
- Once all the WAPs have been patched in at a cabinet, the remaining switch ports should be re-patched using the existing patch cords. The installer shall ensure that all leads that were previously patched to a switch port are reconnected.
- In some cases the number of patched ports may exceeds the number of switch ports available. The installer will be required to rationalise the patching to meet the new switch port capacity.
- Where the number of school devices exceeds the number of switch ports available it will be the school's responsibility to highlight devices for disconnection. Installer to advise school authorised point of contact.
- Ensure the school router is connected to the core switch and power up the router(s).
- Once all switches have been mounted and patched the installer must then complete the "SWITCH COMMISSIONING PROCEDURE"

SWITCH COMMISSIONING PROCEDURE:

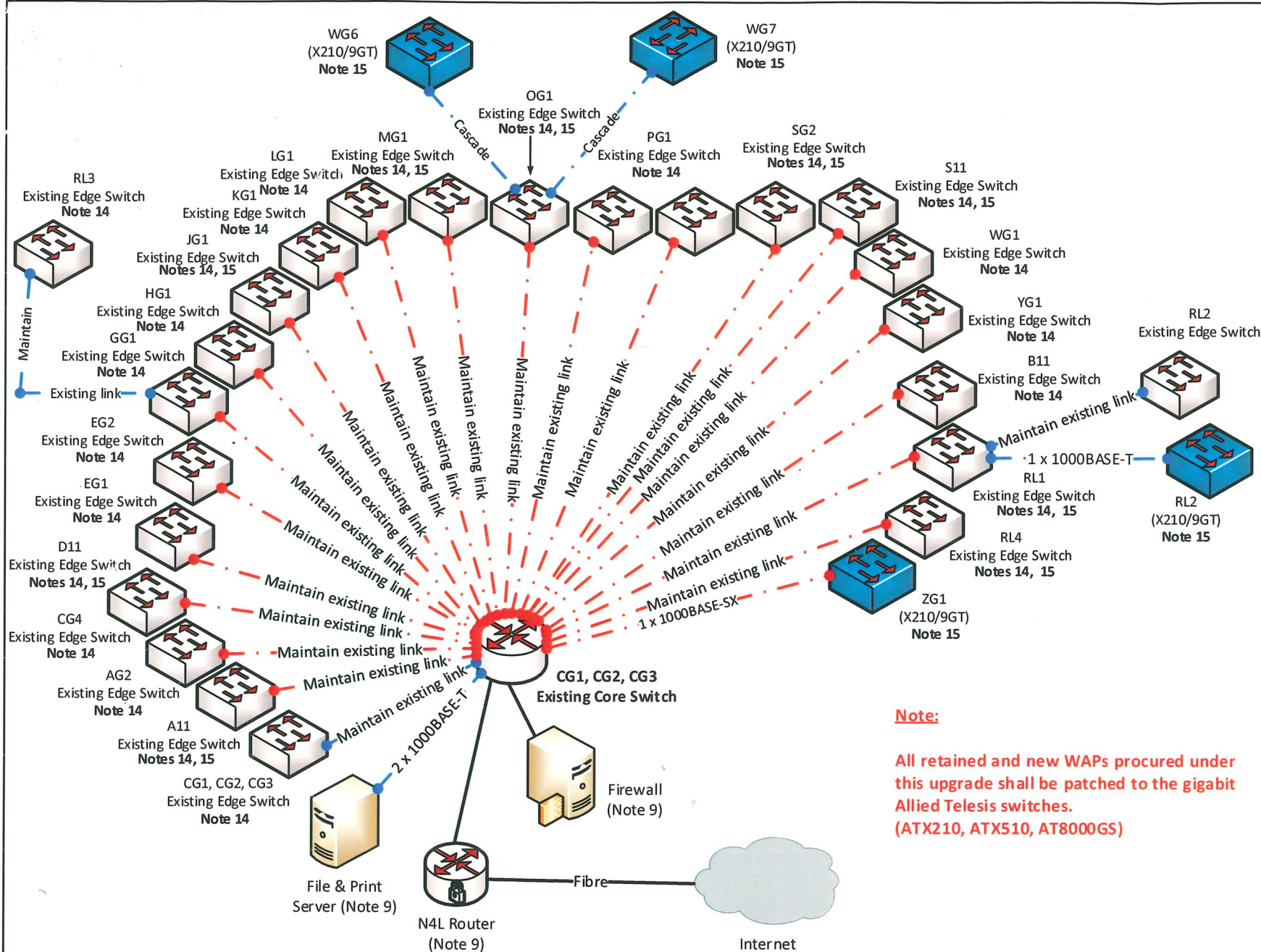
(ONLY REQUIRED IF NEW SWITCHES INSTALLED)

- Request school desktop/laptop from school authorised contact.
- The device must be powered on and physically connected to the school network via a patch cable.
- Ensure the school router is connected to the core switch and power up the router.
- Install Team Viewer on the school device by downloading the setup file from <http://www.teamviewer.com/>
- Execute the setup file and complete a 'Basic installation' for 'non-commercial use'. Once the install is complete Team Viewer should automatically start up and appear on the desktop.
- Call Allied Telesis (0800 114 141) and provide the commissioning engineer with your Team Viewer ID (9 digit number) and password (4 digit number)
- At this point the engineer should have access to the school network via the school device and will undertake the necessary commissioning and testing of the network. The installer may commence the WAP installation once the engineer has indicated they have access to the network.
- Once the switch commissioning has been successfully completed Allied Telesis will inform the installer.
- The installer will need to utilise the desktop/laptop to provide remote access to the Wireless Integrator in a later step. Switch install and commissioning must be completed successfully before the Wireless Commissioning can take place.

HANDOVER DOCUMENTATION REQUIREMENTS

- Prior to leaving site the installer is to ensure all necessary documentation has been gathered. This includes the follow items:
 - Installation Sign off: This document must be signed by the school representative to confirm all hardware has been correctly installed and accepted by the school. NB: This represents the transfer of responsibility of all hardware from the installer to the school.
 - Site Photos: Full view, high resolution cabinet photos (front and back) shall be taken at each cabinet before commencing any WSNUP work. After the completion of the upgrade, photos shall again be taken at all cabinets (front and back) showing all switches and peripherals tidily mounted and patched. Photos shall in addition be taken of all WAPs showing any installed trunking and the TO in a single photo. Failure to provide all photos may result in the installer returning to site to gather this information.
 - Hardware Installation Schedule: Detailing installation and patching information for all access points and switches installed.
 - Test Results: Test Results for all tested TOs shall be provided in both raw and PDF format. Where TOs have been repaired both failed and passed results shall be provided as proof of rectification works completed.
 - The required H&S forms should be completed and signed by the school.
- All documentation shall be uploaded to Torque IP Project Portal (TIPP) within 24 hours of leaving site. Photos should be labelled with their corresponding cabinet/WAP IDs and the Installation Schedule should be provided in excel format.

General Notes, Switch Mounting & Commissioning Procedure and Handover Documentation Notes	Designer: MaryAnne Malaluan		
	SIZE	Release Date	
	A3	12 April 2017	
	SCALE	N/A	
	DWG NO		REV
Contact: Carla Smith	CHS-0340-1606-2450-02-2		0.2
	Job No. 1606-2450	SHEET	2 of 10



Note:
All retained and new WAPs procured under this upgrade shall be patched to the gigabit Allied Telesis switches. (ATX210, ATX510, AT8000GS)

NETWORK SWITCH INSTALLATION REQUIREMENTS

General Notes (All Notes Apply)

01. This schematic provided an overview of the Logical Network including all switches.
02. All works to be undertaken in line with the 'NDP Notes and Procedures' document.
03. The installer is to mount any new switches (Where applicable) and re-patch all existing patch cords. Where the number of patched T/Os exceeds the number of new switch ports provided the installer shall audit the site and remove switch ports connections to any outlets not in use. Where the number of devices still exceed the number of switch ports, then, the following minimum shall be ensured: 4 live outlets per classroom, 2 live outlets per office and only enough live ports for existing devices in PC Studys, reception, copy room and library.
04. The Ministry will provide the patch cords required for switch uplinks and WLAN connectivity free issue, via the PM. For all other patching requirements, at the cabinet and work area end, the installer is to utilise existing patch leads.
05. UPLINK: An uplink port is required for linking two devices directly with a standard patch cord as indicated.
06. STACK: A stack port is required for linking two devices directly with a specific cable as indicated.
07. The Installer should ensure that the existing switches (that are not retained by this project) are removed as part of this project and not left connected to the network under any circumstances. The removal of these older switches ensures the quality of the network upon completion of this project.

Installation Notes

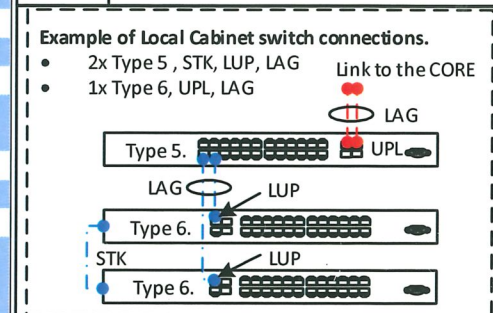
08. Backup link patched to switches at each end, installer is to provide the patch cords in each cabinet to enable the switch integrator to patch the backup links during commissioning, the integrator is to ensure RSTP (or loop back detection for Apple schools) is enabled, prior to connecting this link.
09. Supply of Internet Router, Servers, UPS, Firewall and Active Equipment not listed in the right side of this drawing are not part of the Scope of WSNU. Such equipment should remain in it's current location with all patching relocated to the new switches provided.
10. Existing network switch relocated to this cabinet.
11. The switch at this location shall be cascaded from the previous switch as indicated.
12. If the school have an existing wireless controller, the installer shall first disconnect the old controller before connecting any new wireless access points or the new wireless controller (if applicable) to the school's network.
13. The installer is to ensure that any new Wireless Controller is installed in the Main Cabinet and patched into the core switch.
14. The network switch/stack at this location may not have spare ports, the installer shall un-patch existing WAPs first and re-patch the new WAPs to the same ports. If there are still insufficient ports, the installer shall select a port that is patched to a T/O not currently in use by a device.
15. We are deploying more WAPs in this block than was existing. This site has a WLAN VLAN, any additional VLAN port configuration required shall be conducted by New Era IT prior to wireless cutover.
16. The installer shall patch all WAPs to this switch.

Core Switch Requirements		
Type		Existing
Fibre Links		1
Servers NICs		0
Copper LUP		0
Copper UPL		0
Firewall		0
Router		0
Wireless Controller		0
Fibre Spare		0
Copper Spare		0
Total fibre		1
Total Copper		0
Total ports		1

Cabinet	Switch type	Qty
A11	Existing Edge Switch - AT8000GS/48	1
A11	New PoE Midspan	1
AG2	Existing Edge Switch - AT8000GS/24	1
CG1, CG2, CG3	Existing Core Switch - ATx610/24Ts	1
CG1, CG2, CG3	Existing Core Switch - ATx610/24SPsx (STK)	2
CG1, CG2, CG3	Existing Edge Switch - AT8000GS/48	4
CG1, CG2, CG3	Existing Edge Switch - AT8000GS/24	2
CG4	Existing Edge Switch - AT8000GS/48	1
D11	Existing Edge Switch - AT8000GS/48	1
D11	New PoE Midspan	1
EG1	Existing Edge Switch - AT8000GS/24	1
EG2	Existing Edge Switch - AT8000GS/24	1
GG1	Existing Edge Switch - AT8000GS/24	1
HG1	Existing Edge Switch - AT8000GS/48	1
JG1	Existing Edge Switch - AT8000GS/24	1
KG1	Existing Edge Switch - AT8000GS/48	1
LG1	Existing Edge Switch - AT8000GS/48	1
MG1	Existing Edge Switch - AT8000GS/48	1
MG1	Existing Edge Switch - SMC 8024L2 (Out of Scope - no warranty extension)	1
OG1	Existing Edge Switch - AT8000GS/48	1
PG1	Existing Edge Switch - AT8000GS/24	1
SG2	Existing Edge Switch - AT8000GS/24	1
S11	Existing Edge Switch - AT8000GS/48	1
S11	Existing Edge Switch - SMC 802462 (Out of Scope - no warranty extension)	1
S11	New PoE Midspan	1
WG1	Existing Edge Switch - AT8000GS/48 (STK)	2
WG1	Existing Edge Switch - AT FS724i (Out of Scope - no warranty extension)	1
YG1	Existing Edge Switch - AT8000GS/48 (STK)	2
B11	Existing Edge Switch - AT8000GS/48	1
B11	New PoE Midspan	1
RL1	Existing Edge Switch - AT8000GS/48	1
RL2	Existing Edge Switch - SMC 8024L2 (Out of Scope - no warranty extension)	1
RL2	New Edge Switch - Type 10 (Copper UPL)	1
RL3	Existing Edge Switch - AT8000GS/24	1
RL4	Existing Edge Switch - AT8000GS/48	1
WG6	New Edge Switch - Type 10 (Copper UPL - Cascade)	1
WG7	New Edge Switch - Type 10 (Copper UPL - Cascade)	1
XG1	Existing Edge Switch - ATx510/28T	1
XG1	Existing Edge Switch - ATx510/52T	1
ZG1	New Edge Switch - Type 10 (UPL)	1

Type 10	Form Factor: 19" Rack Mountable 1RU	4
	Layer: 2	
	10Gb SFP Ports: N/A	
	Ports: 8 * 10/100/1000 RJ45 + 1 * SFP	
SFP	Blocking: Fully non blocking	
	Stacking: N/A	
	SFP-SX	
	SFP-TX	
Mod	SP10-SR	0
VLANs	VLAN Configuration Required?	Yes
ON SITE	On-site Switch Commissioning Required?	No
P INJ (WLAN)	Power Injector. Standard Gig unit for WiFi 802.11n access points. Up to 15.4W power. Supports 802.3af standard.	58
PoE Midspan	Form Factor: 19" Rack Mountable 1RU	4
PoE Midspan	Power over Ethernet Midspan.	
	Ports: 12 (Up to 30W of power on 2-pairs)	
	Throughput Data: 10/100/1000Mbps	
PoE Midspan	IEEE 802.3af / IEEE 802.3at compliant	

Key	
Symbol	Description
	OM3 Fibre (1G)
	Cat 6 Copper uplink
	OM3 Fibre (10G)
	Retained Switch Layer 2 (Schematic)
	Retained Switch Layer 3 (Schematic)
	New Switch Layer 2
	LUP Local Uplink
	UPL Uplink to core
	STK Stacked
	LAG Link Aggregated
	Link aggregation



Location: [0340] Cashmere High School

Contact: Mark Wilson

Title:
Logical Network Schematic

Designer: MaryAnne Malaluan

SIZE	Release Date
A3	5 April 2017
SCALE	
N/A	
DWG NO	
CHS-0340-1606-2450-01-6	
Job No. 4/4	SHEET 6 of 10

Document History				
Drn.	Review	Date	Reason	Rev.
MAM	RS	1606-2450-01-6	Initial Draft	0.1

File Name: [School] In Trials-Ministry of Education #Job #, Rev]

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